

Using CNN to Identify Finger Print Image Representations of Optimization Algorithms

Cyrus Orbell

Corresponding author(s). E-mail(s): corbell@trinity.edu;

Abstract

Metaheuristic optimization algorithms are traditionally analyzed using numerical performance metrics such as convergence speed and solution quality. While effective for benchmarking, these methods often fail to capture the structural dynamics of algorithm search behavior within complex search spaces. Recent work has suggested that visual representations of optimization trajectories may reveal additional information regarding exploratory and exploitative search strategies. This paper investigates whether Convolutional Neural Networks (CNNs) can identify metaheuristic optimization algorithms based solely on matrix-based visual fingerprints generated from optimization runs.

Visual fingerprints were generated by applying K-means clustering to optimization trajectories collected from five metaheuristic algorithms executed on functions from the Black-Box Optimization Benchmarking (BBOB) suite. The resulting matrices were converted into 128×128 image representations and used to construct a dataset of 3,600 labeled images. To evaluate generalization, the dataset was divided by benchmark functions rather than random image splitting, ensuring that testing was performed on previously unseen optimization problems. A CNN consisting of three convolutional layers and max-pooling operations was trained to classify the generated fingerprints.

Experimental results demonstrated strong classification performance across the initial five-dimensional benchmark problems, with all algorithms achieving high accuracy and several exceeding 90 percent classification accuracy. Confusion matrix analysis indicated that the CNN successfully learned algorithm-specific visual patterns that generalized across unseen BBOB functions. Additional experiments involving dimensionality transfer from five-dimensional to thirty-dimensional problems revealed substantial differences in how certain algorithms scale across dimensions. In particular, CMA-ES exhibited severe performance degradation under dimensionality shifts, while Differential Evolution and Particle Swarm Optimization maintained relatively high classification accuracy.

The results suggest that matrix-based visual fingerprints contain meaningful structural information regarding optimization behavior and that CNNs can effectively learn discriminative features from these representations. This work

demonstrates the potential of combining visual analysis techniques with deep learning as a novel approach for characterizing and distinguishing metaheuristic optimization algorithms.

1 Introduction

Metaheuristic optimization algorithms are widely used to solve complex search and optimization problems across domains where traditional deterministic methods become computationally infeasible. Algorithms such as Particle Swarm Optimization (PSO), Differential Evolution (DE), Simulated Annealing (SA), and Covariance Matrix Adaptation Evolution Strategy (CMA-ES) exhibit distinct search behaviors characterized by differing balances of exploration and exploitation within a search space. Traditionally, these algorithms are evaluated using numerical performance metrics such as convergence speed, fitness values, or solution accuracy. While effective for benchmarking optimization performance, these approaches often fail to capture the underlying structural dynamics of the search process, particularly in high-dimensional or multimodal problem spaces.

Recent work such as ClustOpt[1] has suggested that visual representations of optimization behavior can reveal information that traditional numerical analysis may overlook. In particular, matrix-based visual fingerprints generated from optimization trajectories can expose spatial patterns associated with candidate solution distributions, clustering behavior, and search-space traversal strategies. As noted in the ClustOpt framework, existing methods “rarely address the location of the solution candidates in the search space, which can reveal crucial information about the exploratory and exploitative strategies of an algorithm.” These representations provide a means of analyzing optimization algorithms not solely through scalar performance outcomes, but through the structural characteristics of their search behavior.

Because these fingerprints are inherently image-based, Convolutional Neural Networks (CNNs) provide a natural framework for automated analysis and classification. CNNs are specifically designed to identify hierarchical spatial features and patterns within images, making them well suited for learning algorithm-specific visual structures embedded within optimization fingerprints. Rather than relying on handcrafted statistical descriptors, a CNN may learn discriminative visual features directly from the generated matrices.

This paper investigates whether a CNN can accurately identify the metaheuristic algorithm responsible for generating a visual fingerprint. The study trains a CNN on image representations derived from multiple optimization algorithms evaluated across different Black-Box Optimization Benchmarking (BBOB) problems. The primary research question is whether algorithm-specific search behaviors produce sufficiently distinct visual signatures for reliable classification by a deep learning model. By exploring this question, the work aims to evaluate the viability of visual fingerprinting as a method for characterizing and distinguishing optimization algorithms.

2 Proposed Methodology

2.1 Optimization Algorithms and Benchmark Problems

This study investigates whether Convolutional Neural Networks (CNNs) can identify metaheuristic optimization algorithms based solely on visual representations of their search behavior. Several metaheuristic algorithms were selected for experimentation due to their relatively simple implementations and distinct search dynamics, providing an initial test of whether algorithm-specific visual fingerprints can be learned by a CNN classifier.

The algorithms were evaluated using the Black-Box Optimization Benchmarking (BBOB) suite, which contains a diverse collection of optimization problems with varying characteristics such as separability, conditioning, and multimodality. A total of 24 BBOB benchmark functions were used. Initial experiments were conducted on five-dimensional problems, followed by additional testing on 30-dimensional versions to examine whether the proposed approach generalizes to higher-dimensional search spaces.

Each optimization algorithm was executed for 100 iterations on every benchmark function. During execution, the positional information of candidate solutions within the search space was recorded for later transformation into visual representations.

2.2 Visual Fingerprint Generation

To generate visual fingerprints of optimization behavior, K-means clustering was applied to the recorded optimization trajectories using five clusters. The clustering process grouped candidate solution positions according to their spatial relationships within the search space, producing matrix-based representations of the search behavior exhibited by each algorithm.

The resulting matrices were converted into image representations using Matplotlib. All generated images were resized to a uniform resolution of 128×128 pixels to ensure consistency during CNN training. These images serve as visual fingerprints intended to capture structural characteristics of the optimization process that may not be evident through traditional numerical performance metrics alone. The final dataset consisted of 3,600 labeled images, with each image associated with the optimization algorithm responsible for generating it. An example of an image is shown in figure 1 where the X-axis represent iterations and the y-axis represents

To reduce overlap between benchmark problems during training and evaluation, the dataset was divided according to BBOB function groups rather than through random image-level splitting. This approach was intended to evaluate whether the CNN could generalize algorithm-specific patterns across unseen optimization problems rather than memorizing characteristics of individual benchmark functions. The dataset was divided into: 14 BBOB functions for training, 5 BBOB functions for validation, five BBOB functions for testing. To ensure diversity across all subsets, each split contained at least one benchmark function from the following BBOB categories: Functions 1–5, Functions 6–9, Functions 10–14, Functions 15–19, Functions 20–24. This stratified grouping was

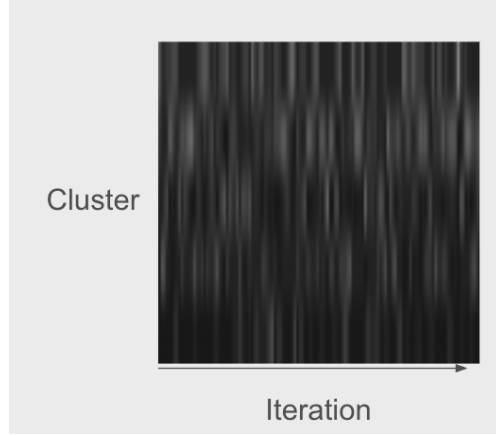


Fig. 1 Fingerprint Representation of an Algorithms

designed to expose the CNN to a broad range of optimization landscapes during both training and evaluation.

2.3 CNN Architecture

The generated fingerprint images were classified using a Convolutional Neural Network (CNN). The network architecture consisted of three convolutional layers, each using a kernel size of 3×3 to extract spatial features from the input images. Max-pooling layers with a pooling size of 2×2 were applied following the convolutional operations to progressively reduce spatial dimensionality while preserving important structural features.

Following feature extraction, the network included a fully connected dense layer containing 128 neurons. Rectified Linear Unit (ReLU) activation functions were used throughout the network to introduce nonlinearity and improve feature learning capability. The final output layer performed multiclass classification corresponding to the optimization algorithms included in the dataset. The model was optimized using the Adam optimizer and trained using a batch size of 32 over 20 epochs.

2.4 Model Evaluation

Model performance was evaluated using both training and validation accuracy across epochs, along with loss measurements to monitor convergence behavior and potential overfitting. In addition, confusion matrices were used to analyze classification performance across individual optimization algorithms and identify patterns of misclassification between algorithms with similar search behaviors.

The primary objective of the evaluation was to determine whether a CNN could successfully learn discriminative visual features from matrix-based optimization fingerprints and generalize algorithm identification across previously unseen BBOB benchmark problems.

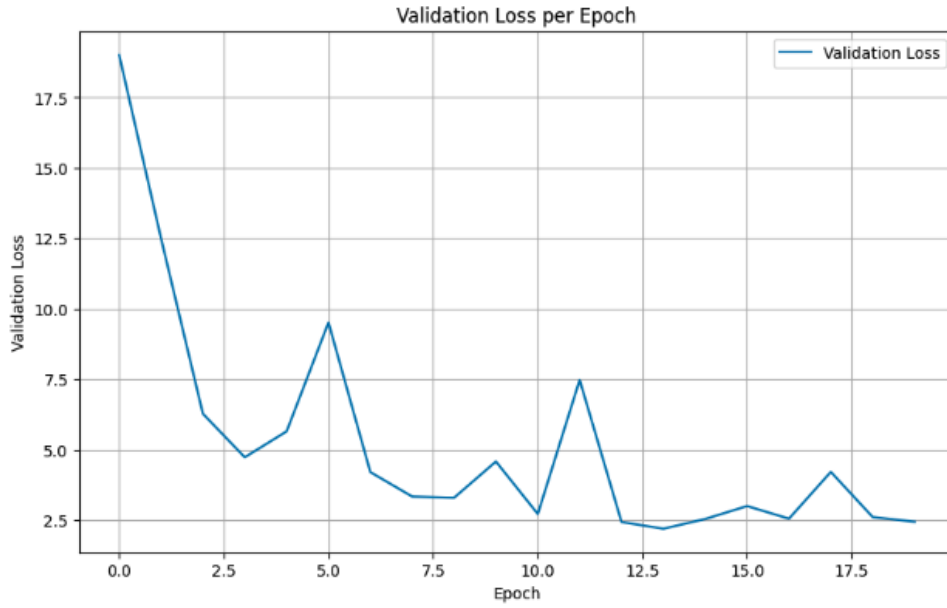


Fig. 2 Validation Loss per Epoch For CNN

3 Results

3.1 Initial Results

The training process of the CNN was successful and showed signs of substantial learning during the training period. Figure 2 shows the model's loss over successive epochs, indicating significant early initial learning followed by continued improvements and refinements as training progressed. The overall trend suggests that the network effectively learned distinguishing features from the training images.

The confusion matrix in Figure 3 represents the results of testing the trained model on a test set consisting of 150 images generated by each algorithm.

The confusion matrix reveals that the model performed exceptionally well across all five algorithms. The lowest performing algorithm, which is PSO, still has an accuracy of above 90 percent, and the incorrect guesses of the model were all very spread apart across the entire board. The highest confusion in all the results was PSO being confused with Simulated annealing. These results reflect the similarity between these two algorithms.

One conclusion that can be made is that the visual fingerprints of the models are able to successfully represent the models. Additionally, the patterns created by the algorithms are recognizable across different BBOB problems as the training database was entirely unique.

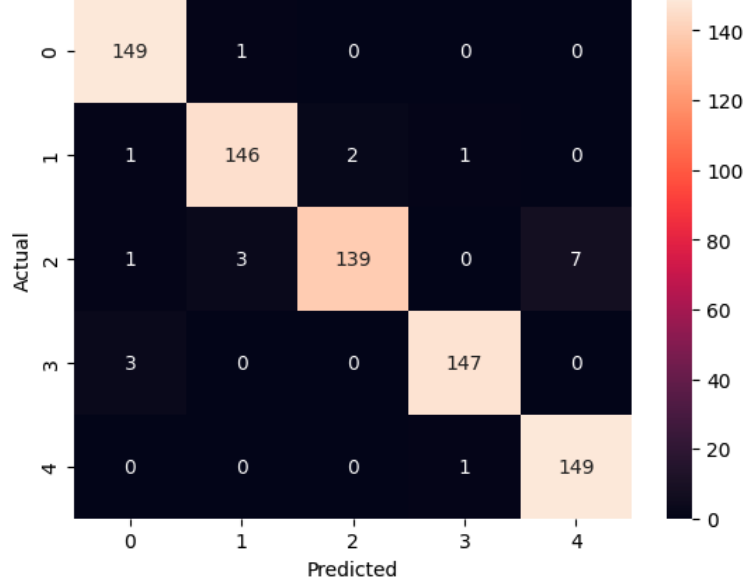


Fig. 3 Confusion Matrix, 1 = CMAES, 2 = DE, 3= PSO, 4 = Random, 5 = SA

The extreme success of these results must also be understood with the knowledge that the model was only trained on five algorithms with each algorithm being substantially unique. With more algorithms and algorithms that are closer related, such as the many nature inspired algorithms, the results would possibly not be as accurate. Also it must be understood that these results were created with a specific size of particles per iteration which may lead to better or worse results if changed.

Despite the caveat the results is still quite impressive and speaks to the power of visual representation and the CNN'S capabilities for recognizing their nuances.

3.2 Further Testing

Beyond the initial experimentation conducted which focused on both training and testing the model on the same dimensionality of problems, I also tested the models effectiveness between dimensionalities.

3.2.1 Setting Up New Training Data

The experimental setup for this phase was similar to the previous configuration, with the addition of a new training dataset generated by running the same five metaheuristic algorithms on the 30-dimensional versions of ten BBOB benchmark problems. The selected functions included f1–f5, representing unimodal optimization problems, and f19–f24, representing multimodal optimization problems. This separation between unimodal and multimodal functions was intended to investigate whether differing landscape characteristics would produce distinguishable variations in the generated visual fingerprints.

3.2.2 New Results

The previously trained five dimensionally model was then tested on the newly constructed data set with the results being saved to the confusion matrices in figures 4 and 5. (Class 0: CMAESClass 1: DEClass 2: PSOCClass 3: RandomSearchClass 4: SimulatedAnnealing)

The results from this testing were significantly less accurate with only DE and PSO maintaining above a 90 percent accuracy and CMAES receiving 0 percent accuracy. Additionally there is little difference in the results between multimodal and unimodal.

The low accuracy in CMAES is a likely indication that the search pattern changes drastically between dimensionalities. The results being 0 percent accuracy as well as simulated annealing being selected 105 incorrectly, indicates that its search pattern is more similar to that of five dimensional Simulated annealing than to its own five dimensional search patterns. This is also something that could be caused by the small amount of classes present in this experiment where the actually most similar algorithm is not in the dataset and thus SA is just chosen as a close second

Conversely DE and PSO had high accuracy both being over 90 percent. This indicates that either these algorithms do not change in a substantial way between dimensionalities or more likely that they do not change in a way that causes them to be more similar to another algorithm in the dataset. Both of these algorithms are relatively simple meaning that they may not be as drastically affected as the other algorithms when changes in dimensionality occur.

The other two algorithms were both quite low. SA saw the most selection by the model but only 30 percent accuracy. This may indicate that the patterns in simulated annealing are similar to the general patterns of 30 dimensional problems or that the model chooses SA as a safe option given some specific visual element being present in the fingerprint. finally random search being around 40 makes since due to the fact that it has no key characteristics making its translation to higher dimensionalities improbable.

Overall the results on the dimensionality shift reveal some key insights into how different algorithms behave as well as how the model may be looking for answers. With further analysis and more datasets more accurate reasoning may be possible.

4 Conclusion

This paper explored whether Convolutional Neural Networks (CNNs) can identify metaheuristic optimization algorithms using matrix-based visual fingerprints generated from optimization trajectories. The results showed that CNNs were able to successfully learn and classify algorithm-specific visual patterns across previously unseen BBOB benchmark problems, demonstrating that these fingerprints contain meaningful structural information about search behavior.

Additional testing across different dimensionalities revealed that some algorithms maintain consistent visual characteristics while others, such as CMA-ES, change significantly as dimensionality increases. These findings suggest that visual fingerprinting may provide insight not only into algorithm identification, but also into how optimization strategies behave across varying search spaces.

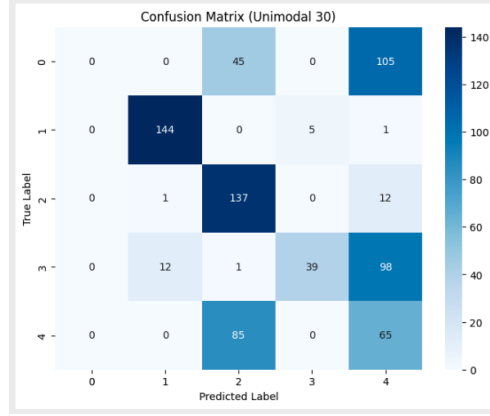


Fig. 4 Unimodal 30D

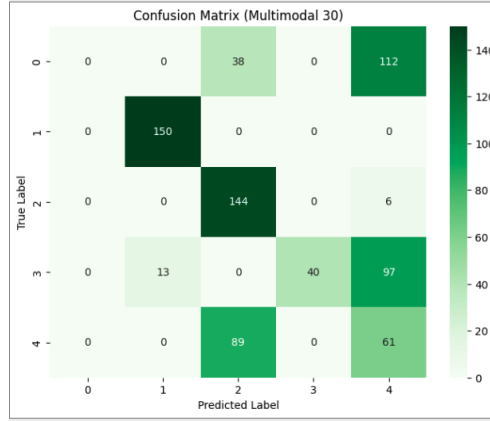


Fig. 5 Multimodal 30D

Although the study was limited to five relatively distinct algorithms, the results demonstrate the potential of combining visual representations with deep learning for analyzing and characterizing metaheuristic optimization algorithms. Future work could expand this approach to larger sets of more closely related algorithms and explore alternative fingerprint generation methods and deep learning architectures.

References

- [1] Cenikj, G., Petelin, G., Eftimov, T.: ClustOpt: A Clustering-based Approach for Representing and Visualizing the Search Dynamics of Numerical Metaheuristic Optimization Algorithms. arXiv preprint (2025) [arXiv:2507.02337](https://arxiv.org/abs/2507.02337) [cs.CV]